

AHMAD BILAL

AI Engineer | Machine Learning | Full-Stack Development

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PROFESSIONAL SUMMARY

Full-Stack AI Engineer with 5+ years of experience developing production-ready AI/ML systems and scalable web applications. Proven track record delivering computer vision, predictive analytics, and enterprise software solutions for industrial clients. Strong foundation in problem-solving and cross-disciplinary collaboration, enhanced by a mechanical engineering background.

TECHNICAL SKILLS

Programming: Python, JavaScript/TypeScript, SQL, HTML/CSS **ML/AI:** TensorFlow, PyTorch, Scikit-learn, Hugging Face, OpenCV **Agentic AI & LLMs:** LangChain, LangGraph, LLM APIs (OpenAI, Anthropic, DeepSeek, Qwen etc.), RAG Pipelines **Data Science:** Pandas, NumPy, SciPy, XGBoost, LightGBM, Joblib, Matplotlib, Plotly **Web/APIs:** Next.js, FastAPI, React.js, Strapi, Node.js **Cloud/DevOps:** AWS, Azure, GitHub Actions, Vercel, Railway

PROFESSIONAL EXPERIENCE

Full-Stack AI Engineer (Contract)

Feb 2025 – Jan 2026

Totalogix SMC Pvt Ltd

Project 1: Gamified Safety LMS | Oct 2025 – Jan 2026 | Next.js, Azure SQL, Azure

- Architected full-stack pnpm monorepo with 10 game types for enterprise safety training; built admin portal with curriculum & user management, analytics export and achievement system
- Deployed to Azure cloud infrastructure; currently serving ~500 users in production

Project 2: Computer Vision Safety System | Jun 2025 – Aug 2025 | Python, PyTorch, YOLOv8, Modbus

- Fine-tuned YOLOv8 for real-time human detection around industrial machinery with ROI masking and live inference pipeline
- Integrated via Modbus to PLC for automated machine shutdown and alarm triggering; deployed on custom Ubuntu workstation, running in factory production

Project 3: Predictive Maintenance System | Feb 2025 – Apr 2025 | Python, XGBoost, PostgreSQL

- Built automated ETL pipeline with degradation-aware feature engineering (thermal drift, vibration escalation, RPM variability)
- Selected XGBoost via time-aware cross-validation achieving 0.87 recall (22% improvement); daily batch inference running in production

AI Engineer & Full-Stack Developer (Freelance)

Jan 2021 – Present

Independent Consultant

- Built NLP apps for document processing and chatbots; created predictive analytics dashboards and ML models for Business Intelligence
- Developed and deployed Next.js/Strapi web applications for clients in transportation and logistics sectors

Research Assistant

Sep 2017 – Sep 2022

DSC Lab, UM-SJTU Joint Institute

- Conducted advanced research in Real-time 3D Object Detection using LIDAR Point Clouds; developed robust CNN-based models for 3D data processing and classification
- Utilized TensorFlow and PyTorch for deep learning model development; achieved computation optimization through CUDA parallel computing techniques

Engineer Intern

Feb 2019 – May 2019

SUPERG.AI, Shanghai

- Worked on autonomous vehicle project — object detection and tracking using LIDAR technology
- Implemented bounding box formation algorithms for detected objects in point cloud data

EDUCATION

M.S. Mechanical Engineering

2017–2022

University of Michigan – Shanghai Jiao Tong Joint Institute

CGPA: 3.5/4.0 | **Thesis:** Real-time 3D Object Detection and Classification using LIDAR Point Clouds

B.E. Mechanical Engineering

2012–2016

National University of Sciences and Technology (NUST)

CGPA: 3.14/4.0 | **Achievement:** Rector's Gold Medal — Best Final Year Project

CERTIFICATIONS

Deep Learning Specialization

Andrew Ng, Coursera

Machine Learning Specialization

Andrew Ng, Stanford

Hugging Face NLP Course

Hugging Face Team

Data Scientist Professional

Python Career Track, DataCamp

Fullstack Django Web Development

Kalob Taulien, Udemy